

Career MAP Program Analysis Draft Report

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1. Executive Summary

This report presents an integrated analysis of the Career Mobility Action Plan (Career MAP) program's two core components, rental assistance and career advancement. The program is designed to reduce housing instability among clients in Washington DC, while supporting them in building sustainable career pathways with higher wages.. The findings reveal important insights into the program and where improvements could strengthen outcomes.

Key Findings

Rental Assistance Analysis:

- Relocation is often associated with higher rents. 86% of program clients ended up with higher rents after relocating, with the typical rent going up from \$1,538 to \$2,676.
- Rent prices vary depending on location. Zip codes like 20019, 20020, and 20032 offer rents \$1,500 - \$1,800 below baseline, while places like 20006 are way more expensive, costing about \$1,400 extra.
- Unit size also play a big part. The higher the number of bedrooms the more the rent.
- Both the number of bedrooms and the neighborhood matter most for how much people pay, 3 bedroom or larger units usually cost \$1,250 to \$2,560 more.
- Many clients pay more in rent than typical market rents for the area especially for 2 and 3 bedroom apartments.

Career Advancement Analysis:

- Several clients earn substantially below market wages in their current roles, particularly business owners, site supervisors, and mail carriers, with observed gaps as high as \$16.48 per hour.
- Most clients aspire to transition into higher-paying fields such as healthcare, information technology, and entrepreneurship.
- Clients are not simply seeking promotions, but rather full career transitions into different fields.

Together, the rental and career advancement results highlight the need for targeted support to help families move into stable housing and higher-paying employment. Strengthening career pathways, supporting credential completion, and guiding families towards more cost-effective housing options can help Career MAP program reduce assistance costs while improving the economic stability and upward mobility of its clients.

2. Introduction

Many public assistance programs, while vital for immediate support, often create an unintended barrier to long-term economic independence through benefit structures that penalize incremental earnings. This phenomenon, commonly known as the “benefits cliff” occurs when a small increase in earned income leads to a sudden, significant loss of critical benefits. Families affected by this face difficult trade-offs between economic advancement and basic stability.

The Career Mobility Action Plan (Career MAP), a pilot program in the District of Columbia was specifically designed to fill this gap. The program offers an integrated approach to help families to continue growing and reaching their full potential, by providing stable foundation and necessary support for career growth.

This report presents a data-driven evaluation of the two major components of the program: Rental Assistance and Workforce Development. Specifically evaluates factors driving costs increases and identifies growth opportunities for program clients. The findings will provide insights for more efficient and sustainable program design.

Project Objectives

1. Conduct an analysis of employment data to identify key occupations and wage gain opportunities for Career MAP participants.
2. Investigate factors contributing to rising Career MAP rental assistance costs, including participant relocations and rental market dynamics.
3. Provide actionable insights to support DHS in optimizing program outcomes, reducing costs, and improving participant success.

3. Literature Review

Public assistance programs including cash assistance, food support, rental subsidies and employment advancement initiatives are critical for promoting family stability and wellbeing in low-income households and vulnerable populations. These programs directly address the financial hardships that contribute to family stress and instability, including housing insecurity, food insecurity, and unemployment. However, the traditional structure of these programs is frequently criticized for creating dependency by often discouraging people from working and growing in their careers. Most government assistance programs have minimum and maximum eligibility requirements. For instance, [Supplemental Nutrition Assistance Program \(SNAP\)](#) requires gross household monthly income, before any program deductions, to be below 130% of the poverty line, the household net income to be at or below poverty line and for the assets to fall below certain limits: households without a member who is aged 60 or older or has a disability must have assets of \$3,000 or less; households with such a member must have assets of \$4,500 or less (Center on Budget and Policy Priorities, 2019). The National Conference for State Legislatures (2025) notes that as families increase their income, they often face a sudden loss of critical support, which in most cases leaves them financially worse off and often this discourages career advancement.

The concept of the ‘benefits cliff’ is central in economic mobility research, describing the sharp drop in net financial resources that occurs when household income crosses a threshold, leading to the abrupt termination of the benefits (Schaefer & Edin, 2014). In addition, Schaefer & Edin notes that this sudden loss can negate the financial gain from a raise or a new job, effectively trapping families in a low-income cycle Altig, Ilin, Ruder, & Terry (2020) also notes that “Some low income workers, particularly those with children, face a disincentive to pursue a higher paying job through so called benefits cliffs, which arise when earnings gains are offset by the loss of means tested public financial supports.”

Numerous studies have confirmed this structural flaw. A key study by the Federal Reserve Bank of Atlanta specifically highlights how benefit cliffs undermine the financial incentives for career advancement, concluding that a gradual phase-out of benefits is essential to ensure that every dollar earned results in a net positive gain for the family (Altig et al., 2020).

The Career MAP program directly aligns with the literature advocating for provision of a comprehensive package of supports, including rental assistance and career advancement opportunities that gradually phase out as clients’ incomes rise, rather than abruptly. By offsetting the loss of benefits through rent discounts and supplemental cash payments, Career MAP aims to remove the financial disincentives associated with benefit cliffs, enabling families to pursue higher-wage jobs and long-term economic mobility without risking their housing or family stability. Research shows rental assistance reduces hardship and provides for a platform to expand opportunity for low income families (CBPP, 2019). Recommendations for literature on public assistance programs has always emphasized on the need for transitional benefits and holistic support to ensure these

programs truly promote self- sufficiency and upward mobility for low-income families. Career MAP's unique value proposition lies in its attempt to create a "ramp" instead of a "cliff", allowing participants to gradually assume full financial responsibility as their earnings increase.

While public assistance programs are essential safety nets, their long-term sustainability is increasingly challenged by escalating costs (CBPP, 2019). This is particularly true for housing assistance, where rising rental markets place significant strain on program budgets (CBPP, 2019). This report uncovers key drivers of cost increases in rental assistance in the Career MAP program, providing data-driven insights for more efficient and sustainable program design. Secondly, this report will identify key career paths and salary increase opportunities for participants in the Career MAP program. By connecting participants with career pathways that offer a clear trajectory for increased earnings, the program will empower families to gradually increase their income and become self-sufficient, thereby reducing their long-term reliance on public assistance and mitigating the negative effects of the benefit cliff (Anderson, 2022). This approach shifts the focus from merely providing a safety net to building a sustainable ladder out of poverty.

4. Methodology

4.1 Data Sources and preliminary descriptors

Rent Assistance Data

Career MAP Rent Data	External Rental Data
Program Entry Rents - Baseline for measuring affordability and rent changes 397 Observations and 5 columns Variables used - <i>response_id, zip_code, original_rent_amount</i>	US Department of Housing and Urban Development 489 Observations, 6 columns Variables used: <i>zip_code, efficiency, bedroom_one, bedroom_two, bedroom_three, bedroom_four</i>
Relocation Rents - To analyze rent amounts when clients move to new housing units and how relocation impacts rental burdens 443 observations, 26 columns Variables used: <i>response_id, rent_amount, new_zip_code, proposed_new_unit_size, household_size, utilities_included.</i>	DCHA Maximum Payment Standards - DC Housing Authority maximum rent allowable for an applicable bedroom size unit. 7 observations, 2 columns Variables used: <i>Number of bedrooms, Payment Standard (maximum)</i>
Current Rents - Latest rent and assistance amount for clients for comparison against market rents 492 Observations, 11 columns Variables used: <i>response_id, full_rent, tcp_total_incl_adj_credit, client_rent_amount, housing_unit_city_state_zip</i>	Rental data – Zillow, Redfin, Rentometer - To compare program rents against current market trends - Webscrapping rental websites for maximum rent amounts by unit size.

Table:1

Workforce Development Data

Career Map	
<i>Intake Survey Data</i> • 518 Observations, 30 columns • Variables used: <i>response_id</i>, <i>education_goals</i>, <i>training_completed</i>, <i>desired_career_interests</i>	<i>Training and Education</i> • 107 Observations, 18 columns • Variables used: <i>training_status</i>
<i>Employment</i> • 41 Observations, 17 Columns • Variables used:	<i>Monthly Data (July)</i> • 498 Observations, 29 columns • Variables used:
<i>Market Data</i> O*Net Online	

Table: 2

Handling Missing Data

- Variables and observations with all or 60% of missing values were dropped.
- Standardized values for some responses that were not necessarily missing values but rather responses e.g the career advancement dataset has responses like “NA”, “None”, “Not Applicable”. All such entries were recoded to None for consistency.
- Data specific amputation was also done for critical variables that didn’t have a lot of missing values. Missing values were not imputed unless justified, instead, analyses was restricted to available entries to avoid introducing bias.

Imbalanced Data Handling

- No imbalance-handling techniques such as resampling or class weighting were used on this project because the analysis centered on descriptive statistics, identifying rent patterns, and employment trends rather than training models to predict rare outcomes.
- The emphasis remained on presenting unbiased summaries and visualizations, without introducing artificial adjustments that could distort interpretation or policy-related conclusions.

Standardization

- **Data Types and Formats:**

All datasets were cleaned using `janitor::clean_names()` to make sure the column names were consistent and make it easier to join and transform data.

Dates and categorical variables were recoded into more consistent formats e.g `forcats` package to manage factor variables.

- **Variable Construction:**

Some text fields, especially in the Workforce Development dataset, were standardized by combining similar responses, for instance, “Bachelor’s Degree” and “Bachelors” into single categories.

- Some variables were also changed for easy analysis, for instance in the current rent dataset, `tcp_total_incl_adj_credit` was changed to become `rent_assistance`, `housing_unit_city_state_zip` to `current_zip`. In the workforce data, variables like `wage_gap` were introduced.

- **Category Engineering:**

Primarily for the Workforce Development data, AI was used in some instances to identify values and group them for easier comparison and analysis.

Merging Datasets

- Inner and full joins were used to merge datasets. For example, in the rental analysis, an inner join on `response_id` combined old unit rents and relocation data. Similarly, the employment dataset with 41 observations) was merged with the monthly dataset with 418 observations, to return only participants present in both datasets, ensuring a consistent set of matched observations for analysis.
- All three rent datasets were combined using full joins on `response_id` to handle their many-to-many relationships. The old rents, new rents, and current rent tables were fully joined so that every record from each source was retained.
- After joining, common fields were standardized by renaming variables (`original_unit_rent` to `rent_old`, `new_rent` to `rent_new`) consolidating zip codes into a single column, and unifying rental dataset for analysis.

Variable Construction

- Constructed certain variables to expand analyses.

- In the rental data, inflation variable was added and for the workforce development, variables such as wage gap were added.
- A variable in employment standardized_category was also created to show original position job title and the assigned category.

Analytic and Statistical techniques

- Descriptive analysis
- Multiple Linear Regression
- Simple workforce development data classification

Code embedded

All code used in this analysis is included in this document. Additional code chunks are available in separate Rental Assistance Analysis and Workforce Development Analysis markdown files.

5. Rental Assistance Analysis

5.1 Program Rents and Relocations

Distribution of Old Unit Rents vs Relocation Unit Rents

Old Unit Rents Summary Statistics

```
# A tibble: 1 × 5
  mean median   min   max    sd
<dbl> <dbl> <dbl> <dbl> <dbl>
1 1755.  1538     0 4200  632.
```

New Unit Rents Summary Statistics

```
# A tibble: 1 × 5
  mean median   min   max    sd
<dbl> <dbl> <dbl> <dbl> <dbl>
1 2812.  2676 1003 5944 1024.
```

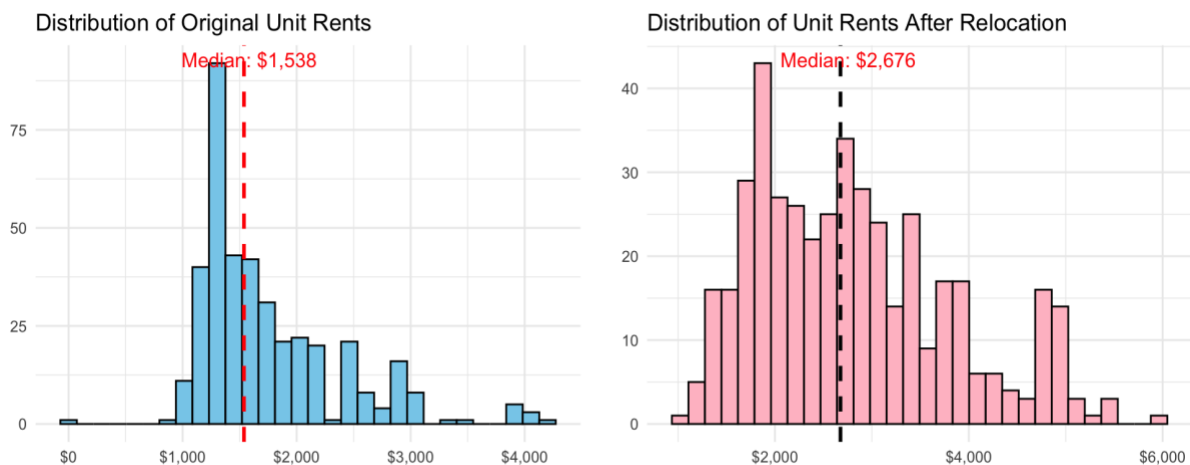


Fig:1

The above histogram shows the distribution of original rents across all units. The majority of units cluster around the media rent of \$1,538. It shows a right skewed distribution with most rents concentrnated around \$1,000 to \$2,000 and most units being below \$2,000. There are outliers on both ends, with some few lower rents and some rent as high as \$4,200.

Rental prices after relocation are distributed across a wider range, with majority of units between \$1,800 and \$4,000, and some as high as \$6,000. There is a jump in the median rent from \$1,538 in older units to \$2,676 after relocation, which is 54% higher. This indicates a significant increase in housing costs after relocation.

Number of program participants in each Zip code before and after relocation.

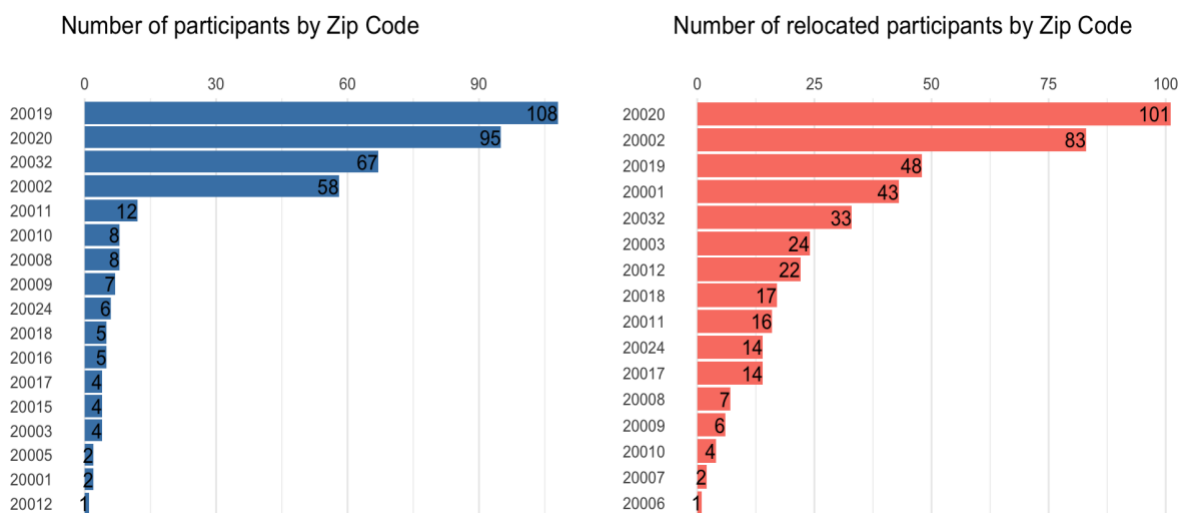


Fig: 2

Most participants lived in zip codes 20019, 20020, 20032 and 20002; neighborhoods east of the Anacostia river and parts of Northeast DC in places including Anacostia, Congress Heights, Bennings and Trinidad. Rents in these areas are usually lower than in other parts of the city.

After relocation, some clients stayed within the same zip code, while others moved to different areas; some affordable, others more expensive. Zip codes 20020, 20002, 20019, and 20001 became the leading destinations for relocated clients.

Geographic heatmap showing median rent across zip codes

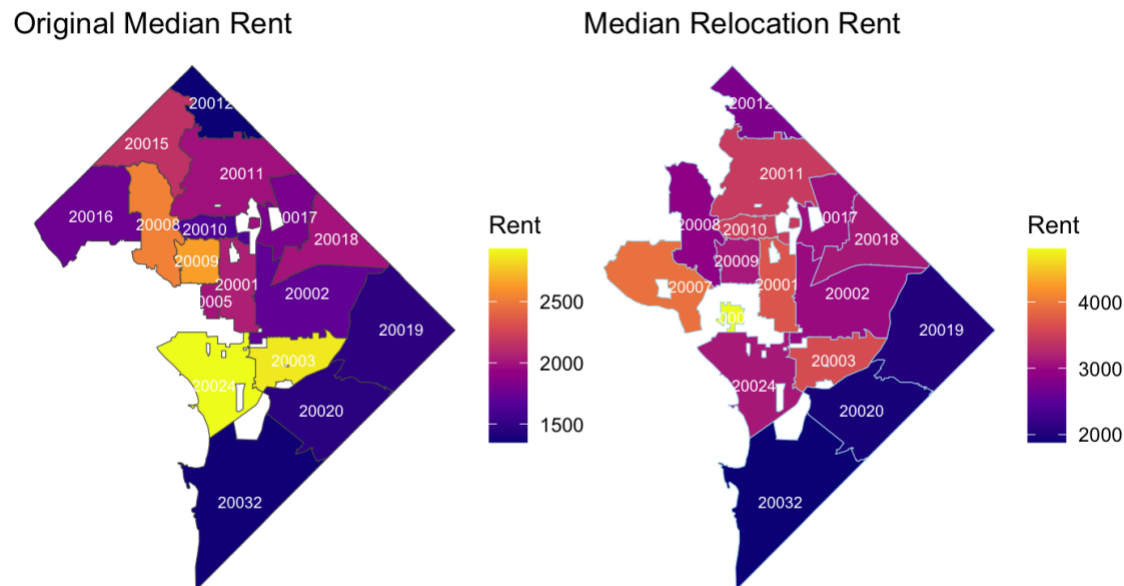


Fig: 3

The geospatial maps compare median rents across DC zip codes before and after relocations. In both maps, the color intensity reflects rent levels, with brighter shades indicating higher rents and darker shades showing relatively more affordable areas. The first map shows where clients were located and second map highlights the areas where they moved to. After relocation, the median rent increased from \$1,538 to \$2,676. Zip codes shown in white are simply places where no program clients lived.

Median rent by unit size overall

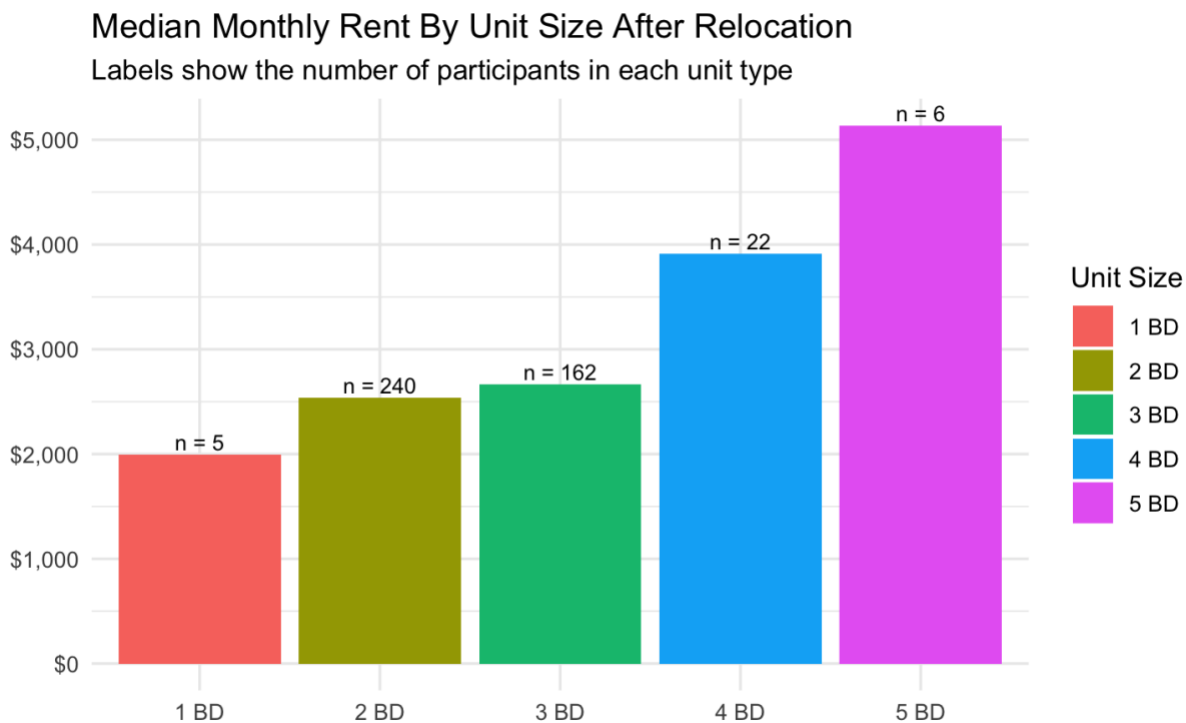


Fig: 4

Rent increases with bedroom size, with larger units obviously attracting higher rents than smaller units. Median for a one bedroom unit is \$1,995, with four and five bedroom units attracting way more than \$3,000 median rent. However, most program clients are concentrated in 2 bedroom (240) and 3 bedroom (162) units, compared to only 5 in 1 bedroom units and 6 in 5 bedroom units.

Key factors driving rent prices - Regression Analysis

The multiple linear regression results show that rent significantly increases with larger unit sizes, 3 bedrooms and above, attracting higher rents than 1 bedroom units. Two bedrooms are associated with an average increase of \$517 in rent, 3 bedrooms add \$1,252, 4 bedrooms add \$2,011 and 5 beds add \$2,561. Larger units (3, 4, 5) seem to be statistically significant compared to smaller units. However, fewer clients occupy larger units, so the program spends less on these sizes.

Several zip codes have statistically significant effects on rent as well, indicating that location plays an important role in rent variation. Household size and inclusion of utilities in rent do not seem to have a significant influence on rent in this model. This model explains about 62% of variation in rent prices, highlighting unit size and location being key drivers of rental costs.

Current Unit Rents Summary Statistics

```
# A tibble: 1 × 5
  mean median   min   max   sd
<dbl> <dbl> <dbl> <dbl> <dbl>
1 2573.  2392.    0  5944 1065.

# A tibble: 1 × 5
  mean median   min   max   sd
<dbl> <dbl> <dbl> <dbl> <dbl>
1  210.   209.    0  2210  165.

# A tibble: 1 × 5
  mean median   min   max   sd
<dbl> <dbl> <dbl> <dbl> <dbl>
1 2264.  2031    0  5705. 1031.
```

Distribution of current unit rents

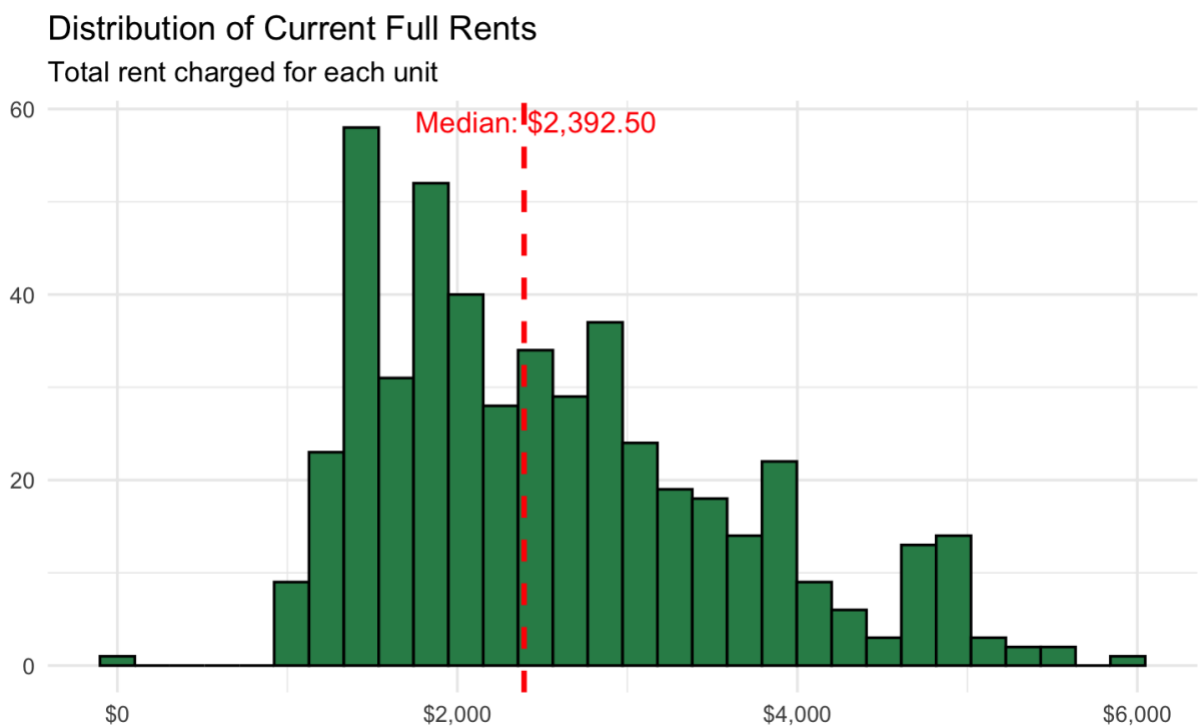


Fig: 5

The histogram shows the distribution of the current full rents across all unit sizes. The rents are generally spread out, with most units falling between about \$1,500 to \$3,500. The median rent is about \$2,393, which is lower than the relocation median rent (\$2,676), but higher than the original unit rents (\$1,538).

Breakdown of full rent, rent assistance, and client share

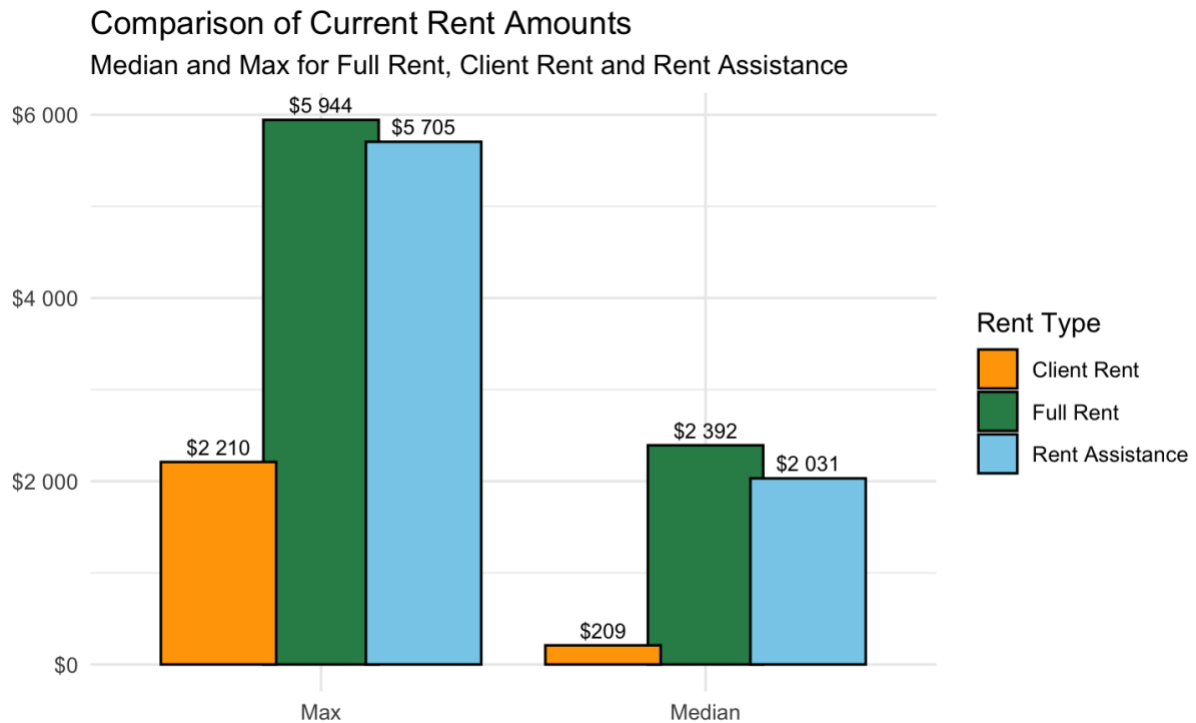


Fig: 6

In the current rent dataset, the highest rent paid for a unit is nearly \$6,000, with a median rent of about \$2,392. The median client rent share is \$209, compared to the \$2,031 subsidy provided through the program.

Rent distribution at different stages

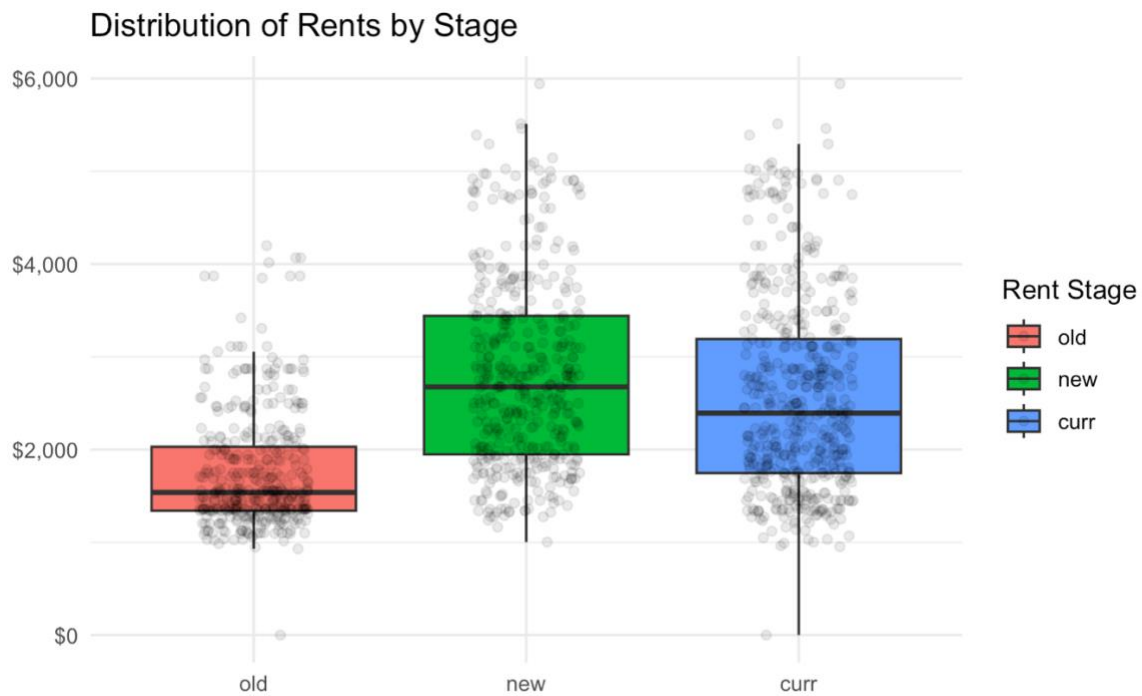


Fig: 7

This plot shows how rents changed for clients before and after relocation, and what they are paying now. Before moving, most rents were lower, as seen in the smaller and tighter range of the “old” rent stage. After relocation, rent amounts jumped higher and spread out with more variation. Current rents remain high but slightly lower than the relocation rents, highlighting a reduction in rent amount but most clients are still paying higher rent than before.

Current Average Rent vs Market Average Rent

Current vs Zillow Average Rent by Zipcode

Top 10 client zip codes by current average rent

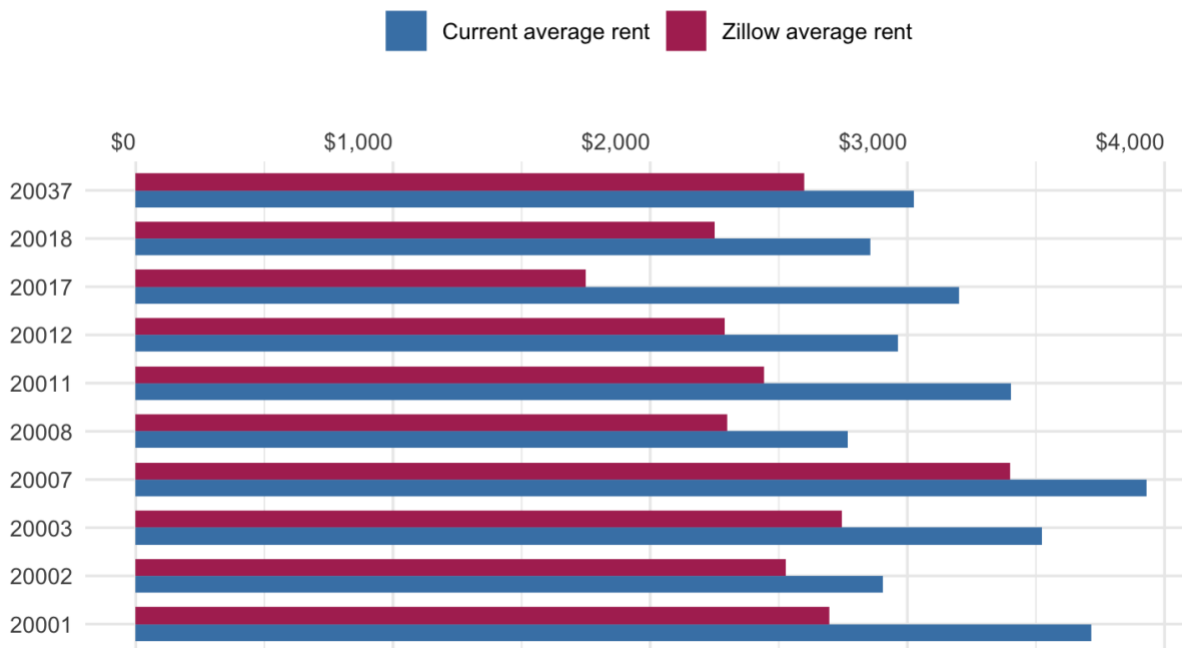


Fig: 8

A comparative analysis of current average rents and market (Zillow) rents by zip code indicates that current rents exceed Zillow estimates in nearly all top zip codes. Some areas show a particularly large gap, suggesting that Career MAP clients may be paying well above market rates in certain locations.

5.2 External Market Data

Fair Market Rents

Each year, the US Department of Housing and Urban Development (HUD) publishes Fair Market Rent (FMRs) estimates to show the typical cost of modest rental housing in different areas of the country. HUD uses these estimates to set payments standards for housing programs such as the [DC Housing Choice Voucher program](#). FMRs help determine how much assistance a household can receive and what the program considers a reasonable rent for that local market. They are based on data about current rents, adjusted for inflation and local market conditions, and are meant to ensure that families using housing assistance have access to safe, decent units without paying more than they can afford (HUD, website). FMR represents the 40th percentile of rents for standard-quality

rental units in a particular area, not the median or maximum, that is, about 40% of rental units are at or below the FMR, and 60% are above it.

An HUD API was obtained to make comparison that would determine whether the program is paying fair market prices or if the rents are higher than average. This data also helps to determine if Career MAP subsidies match recognized benchmarks for housing costs.

Fair Market Rents Summary Statistics

```
# A tibble: 1 × 5  
  mean median   min   max    sd  
  <dbl> <dbl> <int> <int> <dbl>  
1 2482.  2300   840  5000  725.
```

HUD fair market rents by unit size

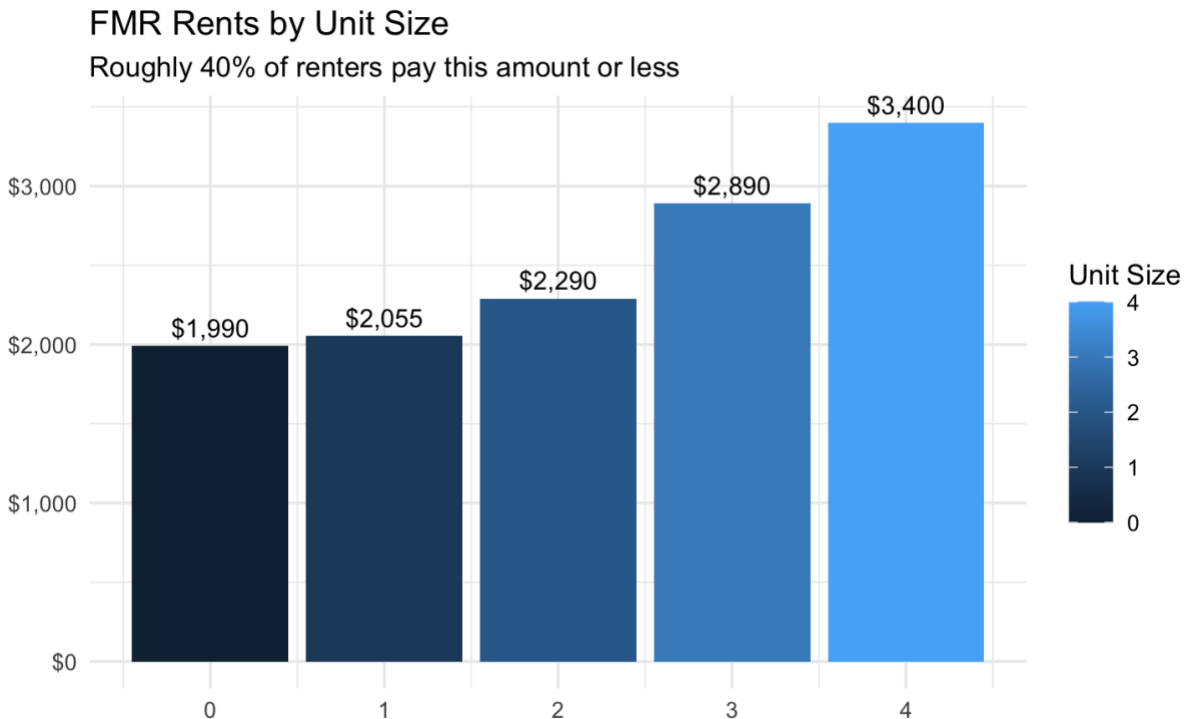


Fig: 9

This plot displays Fair Market Rent (FMR) values for DC as published by the US Department of Housing and Urban Development. Rent increases with unit size with about 40% of families in studios and one bedroom units paying a median of almost \$2,000, while those in two bedrooms pay about \$2,290. Larger units show steep increases, with three bedrooms at \$2,890 and four bedrooms reaching \$3,400.

Median rent by Zip code and Unit size

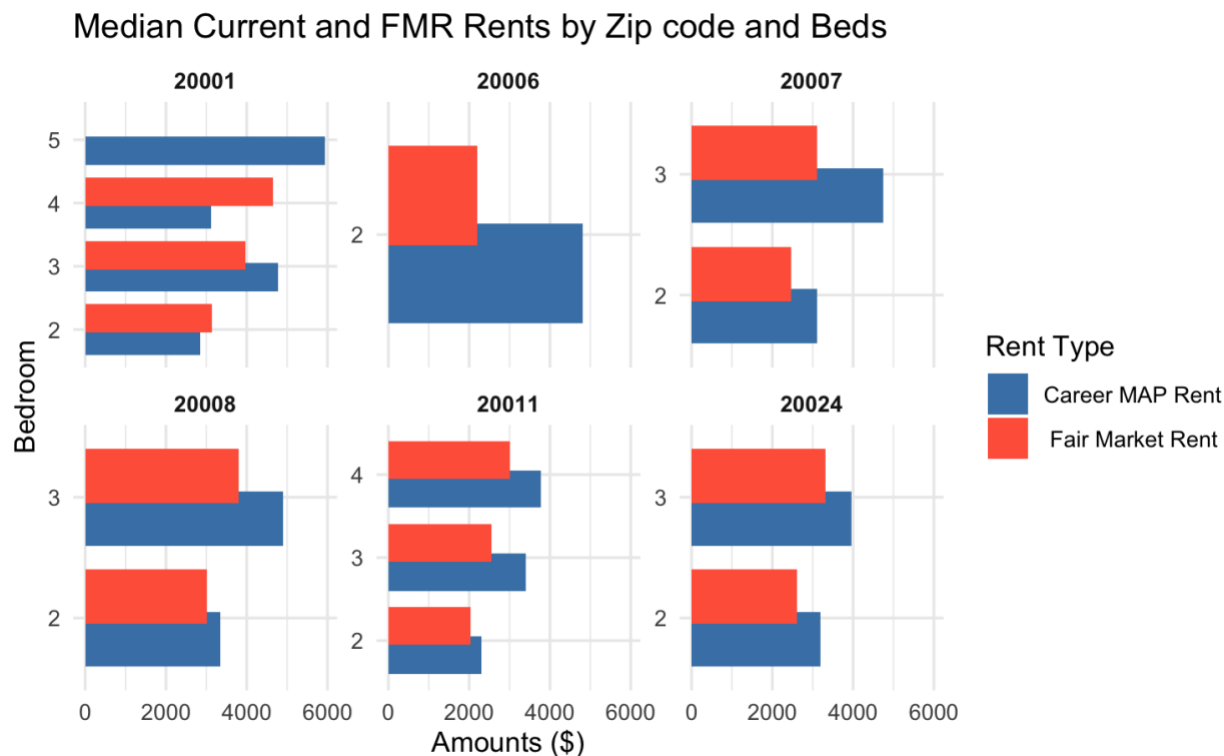


Fig: 10

In most cases, what clients are paying through the program sits above what 40% of renters are paying as shown in the fair market rents. The gap widens as the number of bedrooms goes up, especially in areas like 20006, 20007, and 20024, where larger units push well into the higher price ranges. However, it is more likely that 60% of renters are paying around the same amounts as the program clients.

Other External Sources

Data was manually collected from ZillowRentals' Frequently asked questions. Only the average rent was collected to compare with the average rent in each zip code. More detailed information, such as rents by bedroom size, was not available, as the site listed the same average rents across all zip codes.

Number of beds	Fair Market Rent	Rentometer	Redfin	Full Rent	2025 DC Max Payment Standard	Program Assistance	Count
0	\$2,930	\$1952	-	-	\$3,762	-	-
1	\$3,020	\$2417	\$2287	\$3,277	\$3,845	\$3,114	5
2	\$3,370	\$3360	\$2789	\$4,900	\$4,327	\$4,900	240
3	\$4,250	\$3152	\$2660	\$5,390	\$5,410	\$5,095	162
4	\$5,000	\$4026	-	\$5,008	\$6,382	\$5,008	22
5	-	\$4661	-	\$5,944	\$7,340	\$5,705	6

Table: 3

Redfin, and Rentometer data to compare and get a full picture of the real rental market. Since the current public aggregate data does not offer a more detailed data, this research used median rental prices that were provided in tables and charts of these rental sites.

Data for the comparative analysis was also pulled from the [DC Housing Authority 2025 Maximum Payment Standards](#), which set the maximum allowable amounts for rental assistance in the city. The payment standard reflects 187% of the new HUD Fair Market Rent for all Housing Choice Voucher Programs.

Maximum unit rents in listed zip codes

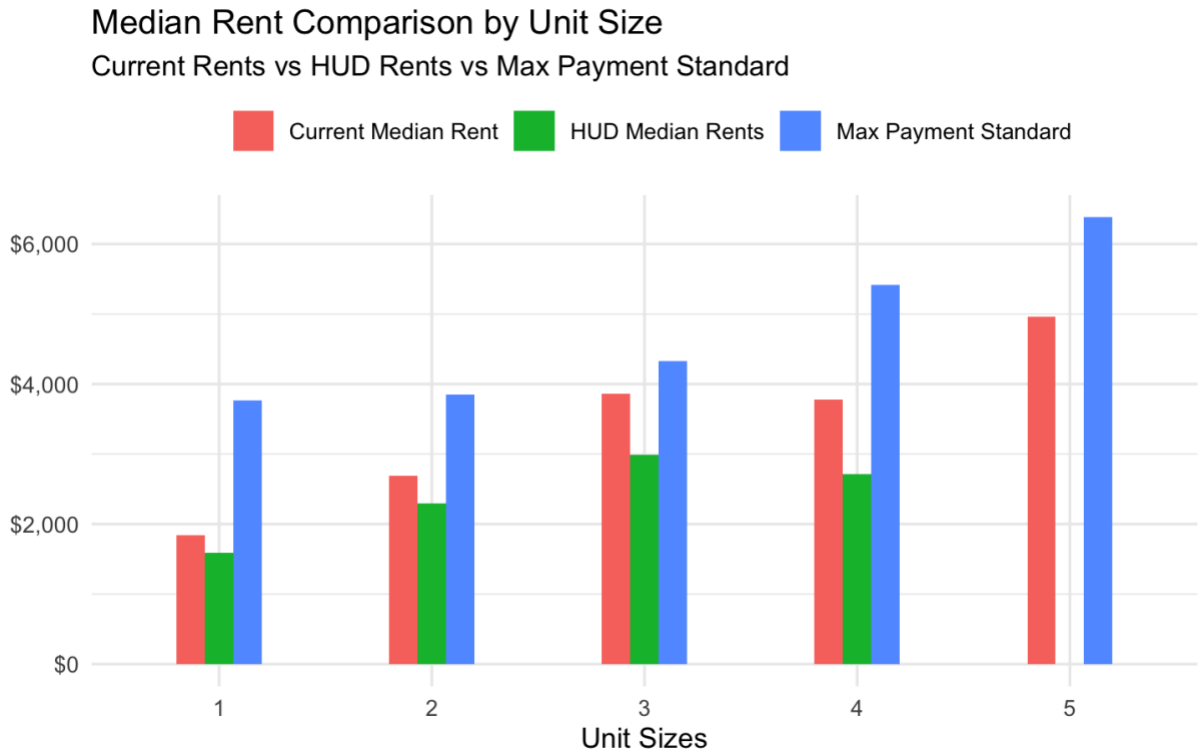


Fig: 11

DCHA also sets maximum payments that are above market rents.

Annual Inflation rate for DC area as of September 2025 was 2.5%.

Current and Predicted Rent by Zip Code

Inflation-adjusted projection using 2.5% annual DC inflation

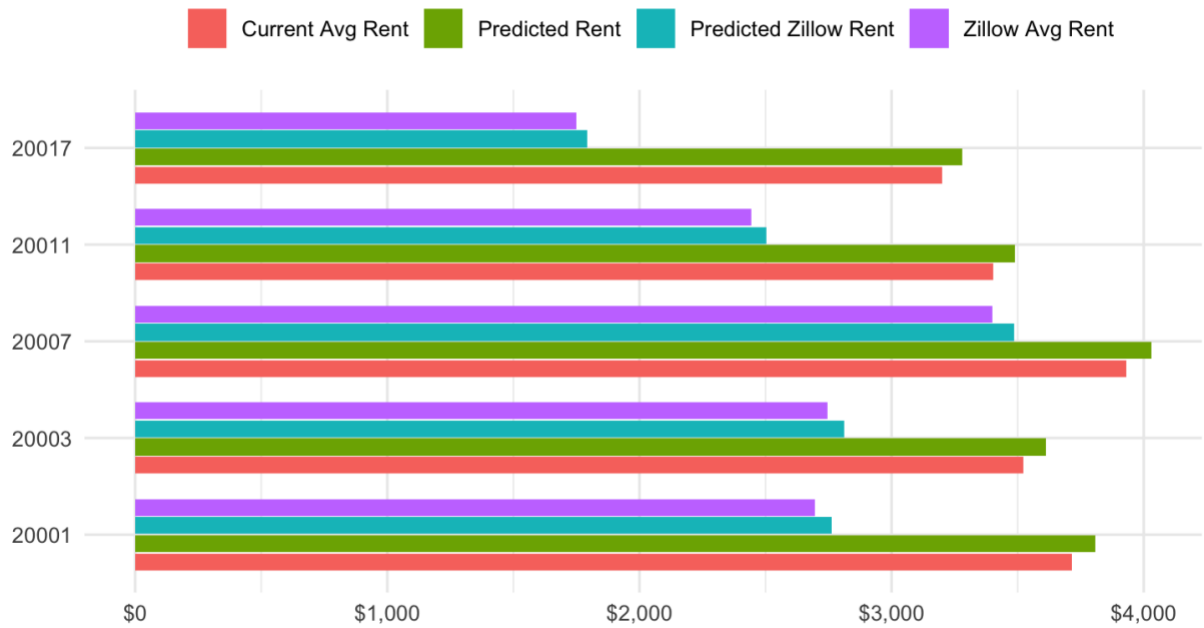


Fig: 12

5.3 Findings

Factors driving increases in rent assistance costs

Since its inception, the DC Career Map program has seen a steady increase in the cost of rental assistance provided to participants. Limited affordable housing, market-driven increases and relocations have all contributed to this rise. [Redfin News](#) reports that median asking rent in Washington DC went up 8.7% in July 2025 compared to last year, the largest jump since September 2022. This increase comes after about two years of mostly stable or falling rents and is largely due to fewer apartments being available while more people are looking for housing (Redfin, 2025). Mobility among clients whether for better job opportunities or improved housing, can further strain the budget due to variable rent levels across neighborhoods.

Relocations

The distribution of rents across the three program datasets highlights relocation as a key factor driving rent increases for program clients in DC. Most rent amounts increased substantially when program clients relocated, with 86% of client relocations resulting in higher rents, and a significant jump in median rent from \$1,538 before relocation to \$2,676 after they relocated. The average rent also increased from \$1,755 to \$2,812, with highest recorded increase after relocation being \$3,865. Relocations were happening within the same zip code or across with some moving from low value to high value neighborhoods thereby increasing the costs and subsequently program costs.

The data also shows a considerable variation in relocation rents, with most units priced between \$1,800 and \$4,000. The range extends from \$1,800 to as high as \$6,000, highlighting significant disparities across different locations and types of units. However, there are some program clients who saw little change and a few who found even lower rents after relocation. Current rents show a slight decrease in rent amounts with a median dropping to \$2,393.

Overall, the analysis shows that moving typically results in higher rent costs compared to remaining in older units, implying that there is limited affordable housing, clients often target more expensive, possibly more desirable or newer properties. The pattern reflects an overall trend often driven by increased housing demand, preferences for higher-quality amenities, better neighborhoods, or increased living space, factors that naturally drive up rents and in turn impacts affordability.

Bigger Units, Higher Rent

A multiple regression analysis highlighted number of bedrooms and zip codes as the major determinants of rent in the data provided, while other variables like household size and

utilities inclusion appear relatively unimportant once other factors are considered. The model showed about 62% of the variation in rent amounts is explained by unit sizes and zip codes. More bedrooms strongly and significantly increases rent amount, units with more than 3 bedrooms having the largest increase and each additional bedroom associated with approximately \$713 higher rent. Larger units are attracting the steepest costs across all rent sources. It is also safer to conclude that, based on the increase in relocation costs, most clients in the program tend to move from smaller units (like studios or one bedrooms) to larger ones as their families grow or their needs change.

Rents Above Market Rates

A comparison of current program rents with DCHA's 2025 payment standards and market data from Zillow, Redfin, and Rentometer shows that, in many cases, program rents are noticeably higher. In most of the top zip codes, current rents exceeded Zillow estimates, with some areas showing large gaps, suggesting that Career MAP clients may be paying above typical market levels.

Looking at rents by bedroom size, 2 and 3 bedroom units were consistently priced higher than comparable market listings. This may reflect high demand as most program participants live in those unit sizes. Many rents also exceed HUD Fair Market Rents (FMR), which represent the 40th percentile and often lag behind real market conditions. However FMR tends to underestimate fast-moving rental prices in DC.

Overall, this pattern suggests that program clients may be paying somewhat above prevailing market rates, likely due to limited availability of lower-cost units and specific housing needs. Rent assistance amounts, and the 2025 DC maximum payment standards—also appear high relative to typical market ceilings.

Future Research Directions on Rental Analysis

This report did not examine several questions that could provide a deeper analysis of the rental data. Additional areas for exploration include:

- Why are clients relocating?
 - Are moves voluntary or forced?
 - Has a change in family size created a need for larger units?
 - Are job changes or relocations influencing geographic moves?
- Does relocation correlate with improved employment outcomes
- Are landlords charging more to program participants or its same rent levels with program clients?

- What is the quality of the units (better amenities, building safety, newer units), and are program clients receiving better housing or simply being overcharged?
- What other factors influence rent variation, given that only 62% is explained by location and unit size?

6. Workforce Development Analysis

6.1 Top Occupations and Hourly Wages

Top 20 highest paying jobs



Fig: 13

Some of the highest paid clients are business owners, entrepreneurs, case managers, customer support representatives, and security guards. Their hourly wages range from \$23 to \$62. Most work full time, while a smaller group is part-time or seasonal. The lowest paid clients include call agent, stocker, front desk, food runners, medical assistants, and suite runners. Their hourly wages typically range from around \$17.10 to \$23 per hour.

6.2 Top Desired Occupations

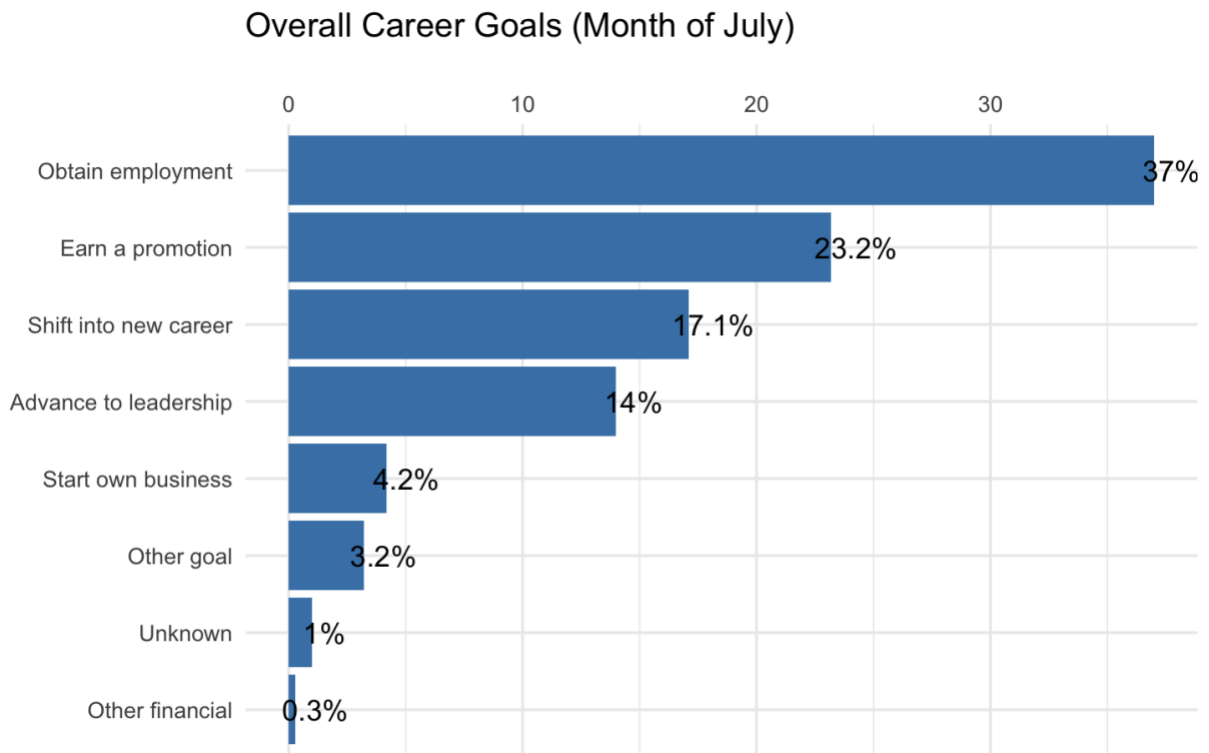


Fig: 14

The plot shows how participants' career goals were distributed in the month of July. The majority of participants' priority is simply obtaining employment, representing a little over a third, while earning a promotion and shifting into a new career are also common but secondary goals. Some few participants want to advance into leadership, start a business or address other financial goals.

Education Goals in July

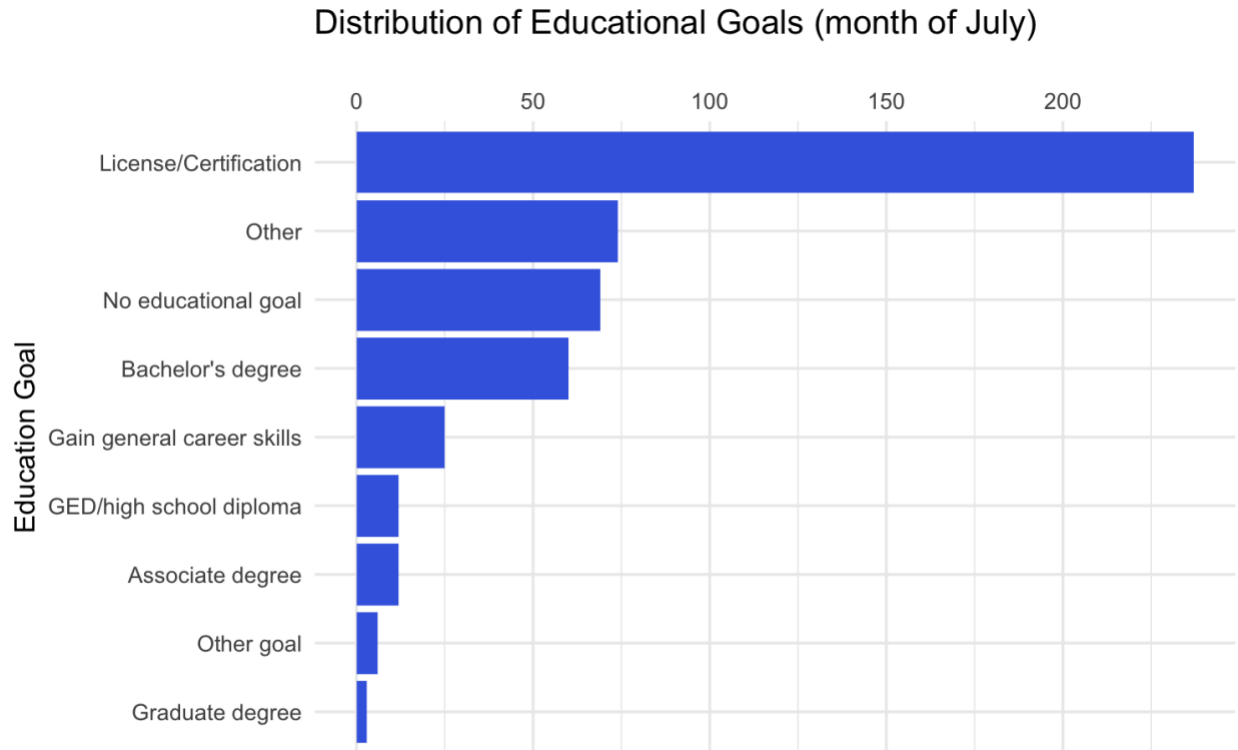


Fig: 15

At the program intake, most program clients had no high school diploma with the next biggest group working towards as associate degree and only a few clients already had higher-level achievements like bachelor's or master's degrees. Some clients did not have any educational goals. However, the monthly data indicate changes in educational goals, with most interested in license or certification, others interested in other goals not that are not educational. At intake, most clients had not completed any formal training. For those who had, most trainings were in healthcare, information technology, customer service, administration, or general office skills.

Career Focus Classification

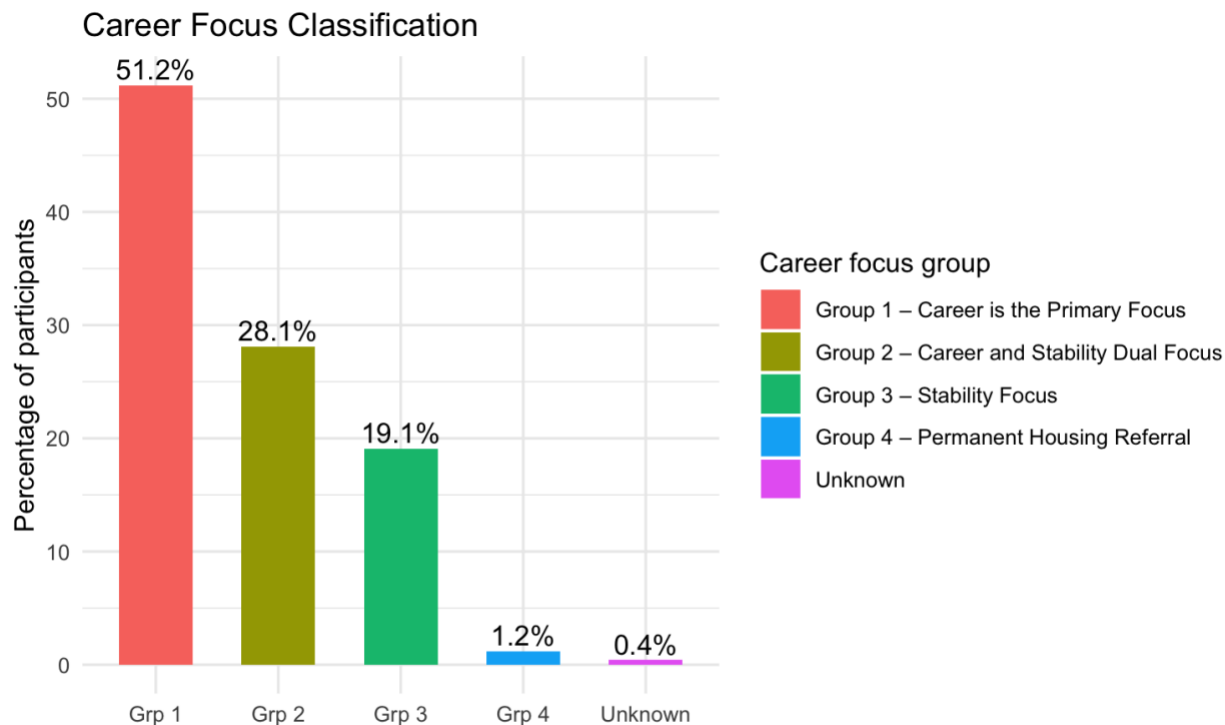


Fig: 16

Some clients cannot focus on workforce goals alone because housing and family stability remain barriers. Over half of the clients' have career as the primary focus. A sizable number fall into dual-focus (Group 2) or stability focus (Group 3) categories, showing that a meaningful portion still need support around stability or housing alongside career help, while only a very small share are in the permanent housing referral or unknown categories.

Desired Career Clusters

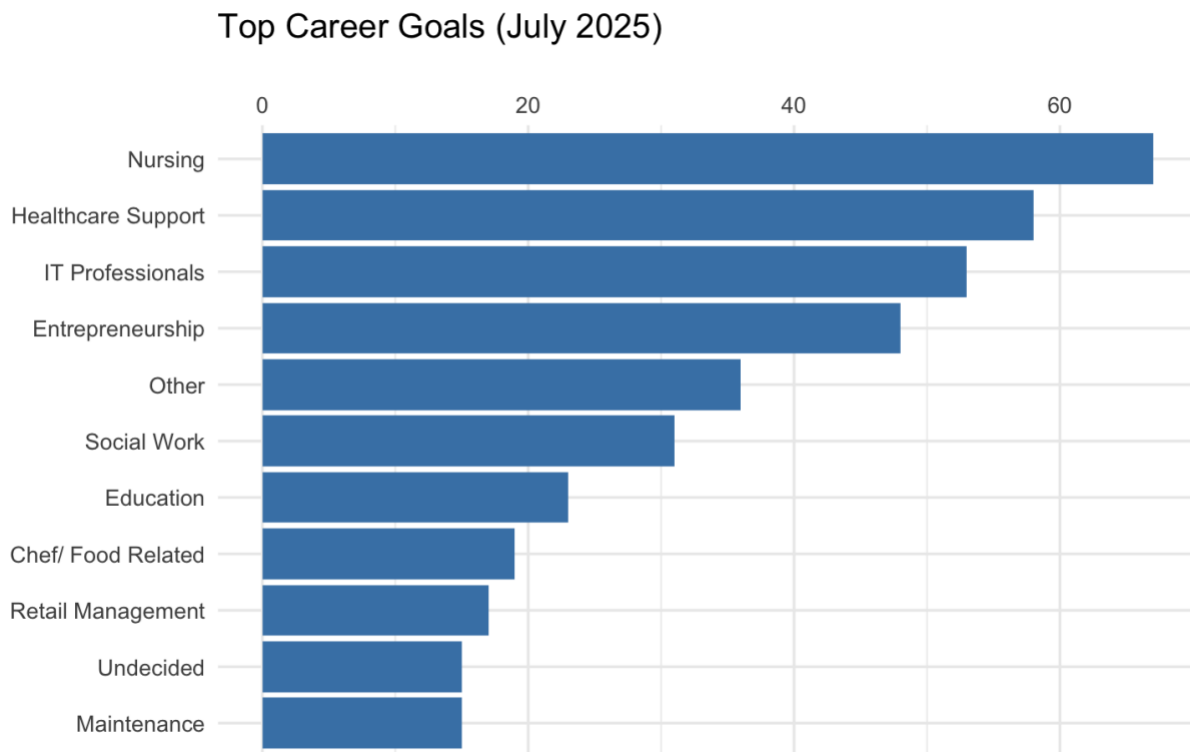


Fig: 17

Most clients are interested in pursuing careers in nursing. Healthcare support roles, information technology, and entrepreneurship also attract significant interest with a smaller number of clients expressing an interest in other industries, and a portion remain undecided about their future career direction.

Current Job Titles Hourly Rates, Market rate and Wage gap

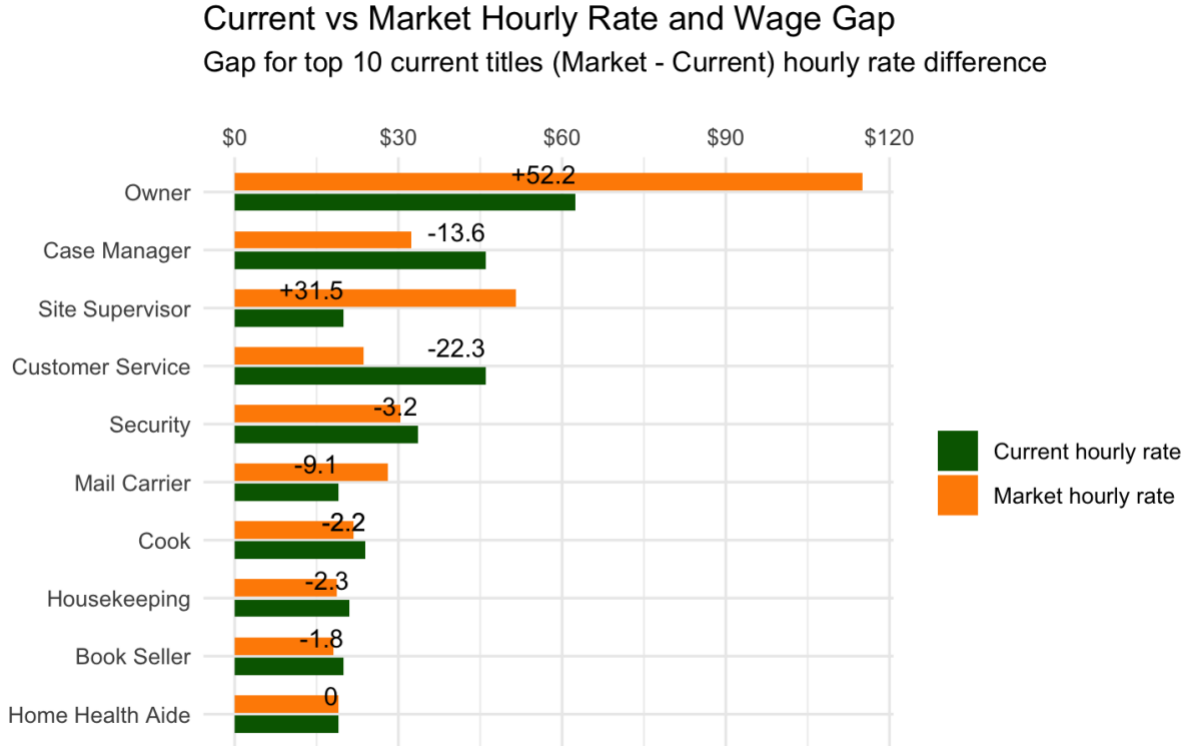


Fig: 18

This plot compares current hourly pay to typical market rates for ten highly paying job titles and highlights where the largest wage gaps exist. Owners and Site Supervisors are seemingly paid far below market levels, In contrast, roles like Case Manager and Customer Service appear to be paid closer to or even above market with smaller or negative gaps. Jobs such as Cook, Housekeeping, Book Seller, Mail Carrier, Security show current pay is fairly aligned with market rates. Some titles have strong potential for higher earnings in the broader market.

Career goals among clients with the highest-paying jobs

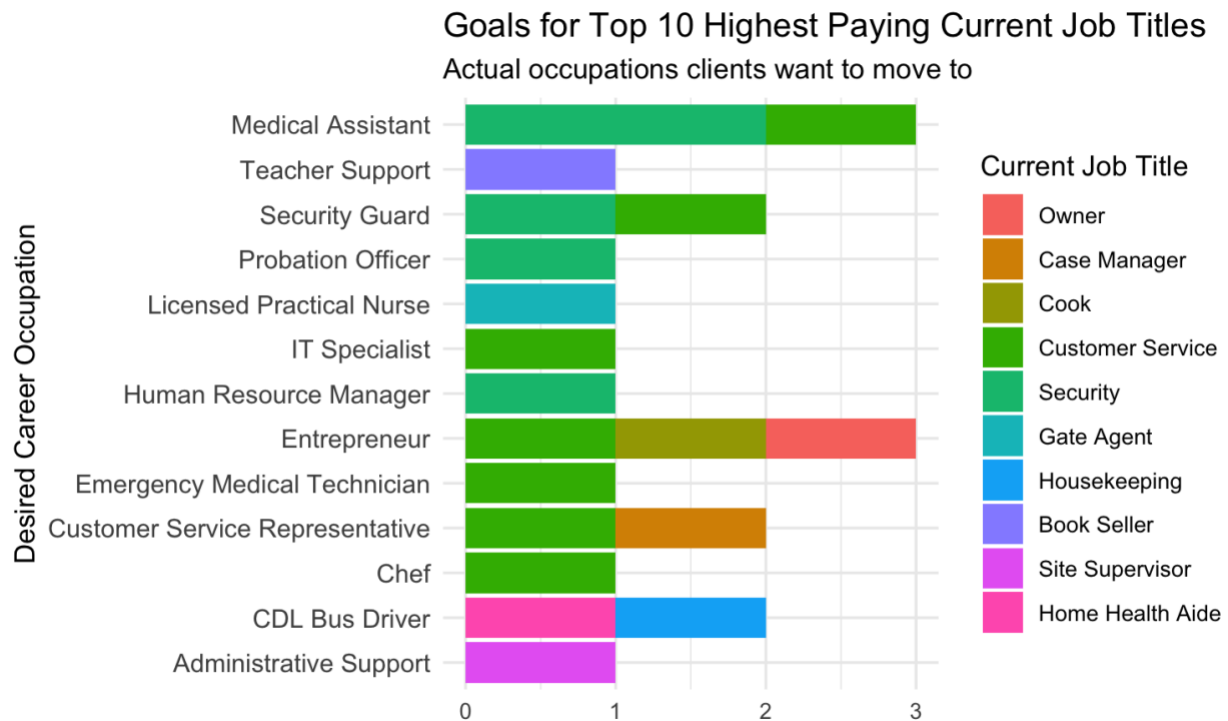
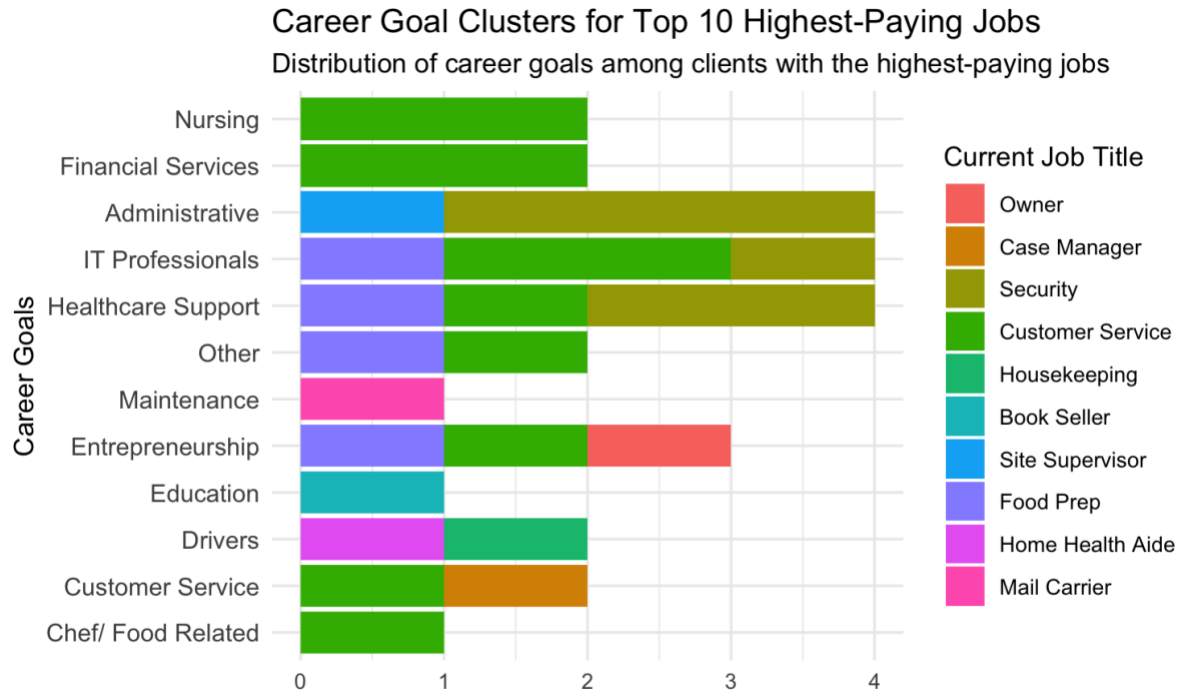


Fig: 19

Many clients hope to move from their current jobs into new career paths in pursuit of better pay and growth opportunities. For example, some clients are aiming to switch from being case managers to customer service representative, from a cook to an entrepreneur, security to a medical assistant or from housekeeping into commercial bus driving. The most desired transitions are into higher-paying fields such as administrative roles, IT and healthcare support.

6.3 Career Growth Opportunities

Current role	Current Wage	High-Mobility Transition Jobs	Education required	Median Wage	Growth Outlook
Owner	\$62.50	Compliance Managers	Bachelor's Degree	\$82.65	Average
		Chief Executives	Master's Degree	\$115.00	Average
Case Manager	\$46	Clinical and Counseling Psychologists	Masters Degree, Post-doctoral training	\$46.07	Much faster than average
		Medical and Health Services Managers	Bachelor's degree, Master's degree	\$77.46	Much faster than average
Security Guard	\$33.56	First-Line Supervisors of Security Workers	Training, High School Diploma	\$37.64	Average
		Security Management Specialists	Bachelor's degree, Vocational Training or on-the-job training	\$51.35	Average
Customer Service Representative	\$23.57	Sales Representatives of Services, Except Advertising, Insurance, Financial Services, and Travel	High School Diploma	\$37.50	Average
		Sales Managers	Bachelor's Degree, on-the-job training or	\$74.19	Faster than average

			vocational training		
Housekeeper	\$21	Lodging Managers Hospitality Manager	Bachelors, on-the-job training, vocational training	\$35.14	Average
		First-Line Supervisors of Personal Service Workers	Vocational training, on-the-job training or Associate Degree	\$27.16	Much faster than average
Book seller	\$20	Library Assistant	High School Diploma	\$25.43	Decline
		File Clerk	Associate Degree/ High School Diploma	\$27.67	Decline
Site Supervisor	20	Solar Energy Installation Managers	High School Diploma, on-the-job training	\$46.31	Faster than average
		Construction Manager	Bachelor's degree	\$61.91	Much faster than average
Food Prep	\$24	Chefs and Head cooks	Vocational training, Apprenticeship or Associate Degree	\$35.58	Much faster than average
		Food Service Director	High School Diploma, Apprenticeship	\$38.58	Faster than average
Home Health Care Aide	\$10	Nursing Assistants	Certification	\$22.53	Slower than average
		Licensed Practical and Licensed Vocational Nurses	Certification	\$33.86	Average
Mail Carrier	\$19	Cargo and Freight Agents	Bachelor's Degree, Apprenticeship	\$23.99	Much faster than average
		Light Truck Drivers	High School Diploma	\$22.89	Much faster than average
		Transportation , Storage, and Distribution	Bachelor's degree,	\$69.32	Faster than average

		Managers			
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Source: US BLS and O*NET Online

Table: 4

The table highlights career pathways for the top 10 most paying roles that program participants currently hold and potential higher-paying and faster-growing jobs they can move into. Across the board, healthcare, management, sales, construction, and energy-related positions show the strongest potential for both increased wages and job growth. For example, case managers and home health aides can significantly boost earnings by moving into healthcare management or licensed nursing roles, while site supervisors and food prep workers can transition into managerial positions in construction, energy, or hospitality. Even workers in lower-wage roles, such as customer service or housekeeping, have opportunities to increase pay through supervisory or specialized positions, though some transitions may require additional education or training. However, occupations like bookselling offer limited growth, with only limited wage increases and declining demand.

Worthy to note is that most people do not want to advance in their careers, they want complete career changes requiring new foundations and trainings.

6.4 Findings

Wage Gap Analysis: Undervaluation

A comparison of current job hourly wages and market wages for the same titles revealed that some program clients are paid way less than the market rate. Owners (likely small businesses struggling to grow), site supervisors and mail carriers are all earning below market rates with the highest wage gap being \$16,48/ hour below market. However, some clients do earn above market rates in their current roles.

This wage gap could suggest that Career MAP clients are working for employers who undervalue their positions, their jobs are in low-income geographic locations or they face other labor market barriers. The gap between credentials and experience could also justify lower pay.

However, it is important to note that most program clients aspire to move to careers with good market rates such as Nursing, IT but currently earn below-market wages in their existing roles suggests that's training programs should also include wage negotiation modules to engage clients to negotiate for high wages in their careers. Most clients are seemingly not interested in advancing in their careers but rather want to change their careers completely.

Desired Occupation and Education Requirement

Higher-paying ambitions, such as entrepreneur or director of food service operations, typically require more advanced education, while other jobs (customer service, chef, bus driver) require vocational training, certificates, or a high school degree. The Career MAP data show many clients gravitating towards opportunities that promise greater financial stability but may be limited by lack of required qualifications and educational barriers. The results point to the importance of continued support and targeted trainings so clients can access career paths that are both attainable and financially rewarding.

Desired Occupations vs Reality Gap

Many program clients aspire to transition into higher-paying fields such as nursing and information technology. For instance, site supervisor to move into administrative support, gate agent to be a Licensed Practical Nurse. However, completely changing careers require more extensive training and advancement within a field, explaining why training completion is relatively low. Wage gap could also be a result of people starting over in new fields at entry level.

7. Limitations

Rental Assistance Analysis

- **Restricted access to key rental datasets:** Inability to access APIs/ important datasets from platforms like Zillow and Redfin, for detailed rent numbers, which limited granular comparison between program rental results with the broader market.
- **Missing rent and program data:**
Career MAP old rent and current rent datasets were missing bedroom sizes, crucial information that would have enhanced analysis and fully reflected real differences or patterns in the data.

Career Advancement Analysis

- **Highly unstructured datasets:** The workforce development datasets needed a lot of cleaning, reorganizing, and standardizing before it could be used for analysis which was time consuming.
- **Multiple responses from the same person:** Clients listed responses in free-text format which required manual categorization. This introduced potential misclassification, especially when titles were vague or overlapped multiple categories.

Other

- Limited time and too many datasets
- Data from different sources needed extra cleaning: Bringing together information from multiple sources required heavy pre-processing, including fixing missing values, matching columns, and aligning definitions.

8. Conclusion

The findings highlight clear opportunities for program improvements. By supporting credential completion, guiding participants towards high-growth, higher-wage occupations, and encouraging relocations to more cost-effective neighborhoods or bulk rent negotiations, the Career MAP program can reduce assistance costs while promoting economic stability and upward mobility for participants.

9. Recommendations

Critical Program Recommendations:

1. Negotiate bulk rental agreements in high-concentration zip codes
2. Strengthen training-to-employment pipelines with employer partnerships
3. Create wage negotiation support to address market rate gaps
4. Develop tiered career pathways (career ladder) with clear milestones from entry-level to target positions.

Data Collection and Processing Recommendations

1. Centralized databases and standardized data formats to facilitate integration across housing, workforce, and rental assistance systems.
2. Data cleaning and management tools to improve data quality, reduce redundancies, and enable faster, more accurate analyses.
3. Consistent template for collecting career, training, and rental information across all programs including use of dropdown menus or standardized occupation codes (eg using Bureau of Labor and Statistics Occupation categories) to reduce free-text job titles that cause misclassification.
4. Data entry to follow uniform data entry rules so that fields like certifications, job goals, and employment outcomes are captured in a predictable way.
5. Agreements with rental platforms, for instance Zillow and Redfin to routinely receive up-to-date data on rent trends by zip code and unit size.

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11. Appendices

Data Dictionary

Rental Assistance Analysis

Relocations_Old Rents.xlsx (old)

Variable Name	Original Field	Description	Processing	Analytic Use
response_id	Response ID	Client identifier	Converted to character/factor	Joining datasets
original_unit_rent	Original Unit Rent	Rent of original unit	Numeric conversion, NAs handled	Comparison with new rents, descriptive stats
original_zip_code	Original Zip code	Zip code of original unit	Extract 5 digits, factor conversion	Grouping, average/median rent by zip, mapping
household_size	Household Size	Number of individuals in household	Not modified	Regression, analysis of rent determinants
utilities_included	Utilities Included	Whether utilities included	Not modified	Regression on rent determinants

Relocations_New Rents.xlsx

Variable Name	Original Field	Description	Processing	Analytic Use
response_id	Response ID	Client identifier	Converted to character/factor	Joining datasets, tracking rent changes
rent_amount	Rent Amount	Proposed new unit rent	Numeric conversion, NAs handled	Comparison with old rents, descriptive stats
new_zip_code	New zip code	Zip code of proposed unit	Extract 5 digits, factor conversion	Grouping, average/median rent by zip, mapping
proposed_new_unit_size	Proposed new unit size	Number of bedrooms of new unit	Factor conversion, NAs handled	Regression, median rent by unit size

rent_change	Derived	Difference between new and old rent	Calculated as new_rent - old_rent	Identify rent increase/decrease
rent_increase	Derived	Logical if rent increased	Derived from rent_change > 0	Proportion of clients with rent increase
same_zip	Derived	Logical if stayed in same zip	Derived from old_zip == new_zip	Classifying move type
move_type	Derived	Stayed or Moved	Derived from same_zip	Mobility analysis

Current Rents

Variable Name	Original Field	Description	Processing	Analytic Use
response_id	Response ID	Client identifier	Character/factor	Joining datasets
full_rent	Full Rent	Total rent for current unit	Numeric conversion	Distribution analysis, comparison with other rents
client_rent_amount	Client Rent Amount	Portion paid by client	Numeric conversion	Analysis of client financial burden
rent_assistance	TCP Total Incl Adj Credit	Portion paid by program/subsidy	Numeric conversion	Analysis of program support
current_zip	Housing Unit City State zip	zip code of current unit	Extract 5 digits, factor conversion	Grouping, average/median rent by zip, mapping

US Housing and Urban Development – Fair Market Rents (DC)

Variable Name	Original Field	Description	Processing	Analytic Use
market_rent	HUD FMR	Fair Market Rent by Zip and bedroom size	Filtered by ZIP, reshaped to long format	Comparison with new rents, regression, visualization
zip_code	Zip	Zip code of unit	Factor conversion	Grouping, mapping

bedrooms	Bedroom Type	Number of bedrooms for FMR	Mapped from HUD field	Comparison of FMR by unit size
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Intake Data

Variable Name	Original Field	Description	Cleaning / Processing	Analytic Use
response_id	Response ID	Client identifier	Not modified	Used as key for merging datasets
education_goals	Education goals	Desired level of educational attainment	Normalized into categories using keyword matching	Used to assess educational aspirations
training_completed	Training completed	Training or certifications participants completed	Mapped into industry categories	Used to evaluate completion status
desired_career_interests	Desired career interests	Participant-stated career interests	Keyword classified into major career sectors	Used to identify career trends
education_goals_clean	Derived	Normalized education level	Generated via regex classification	Used for summary statistics
training_category	Derived	Industry grouping of completed training	Regex mapping	Used for categorical distribution of plots
desired_careers	Derived	Career interest category	Regex mapping	Used for priority workforce areas

Employment Dataset (Employment Data_AU2.xlsx)

Variable Name	Original Field	Description	Processing	Analytic Use
response_id	Response ID	Client identifier	Retained	Joined to July data
position_job_title	Job Title	Current employment	Title case + normalization	Salary comparison

job_classification	Job classification	Employment category- fulltime/ partime	Retained	Workforce segmentation
new_wage_amount_hourly	Hourly wage	Wage at program entry	Filtered for NA	Wage analysis
mean_wage	Derived	Average wage by title	Grouped & summarized	Wage analysis
n	Derived	Job count per title	Calculated	top job rankings

July monthly

Variable Name	Original Field	Processing	Analytic Use
response_id	Response ID	Retained	Inner and full join key
educational_goal	Educational Goal	Recoded into discrete categories	Progress tracking
educational_goal_status	Status	Standardized labels	Progress evaluation
career_goal_occupation_title	Career aspiration	Text cleaned	Career mapping
career_focus_classification	Career focus	Grouped into four levels	Program segmentation

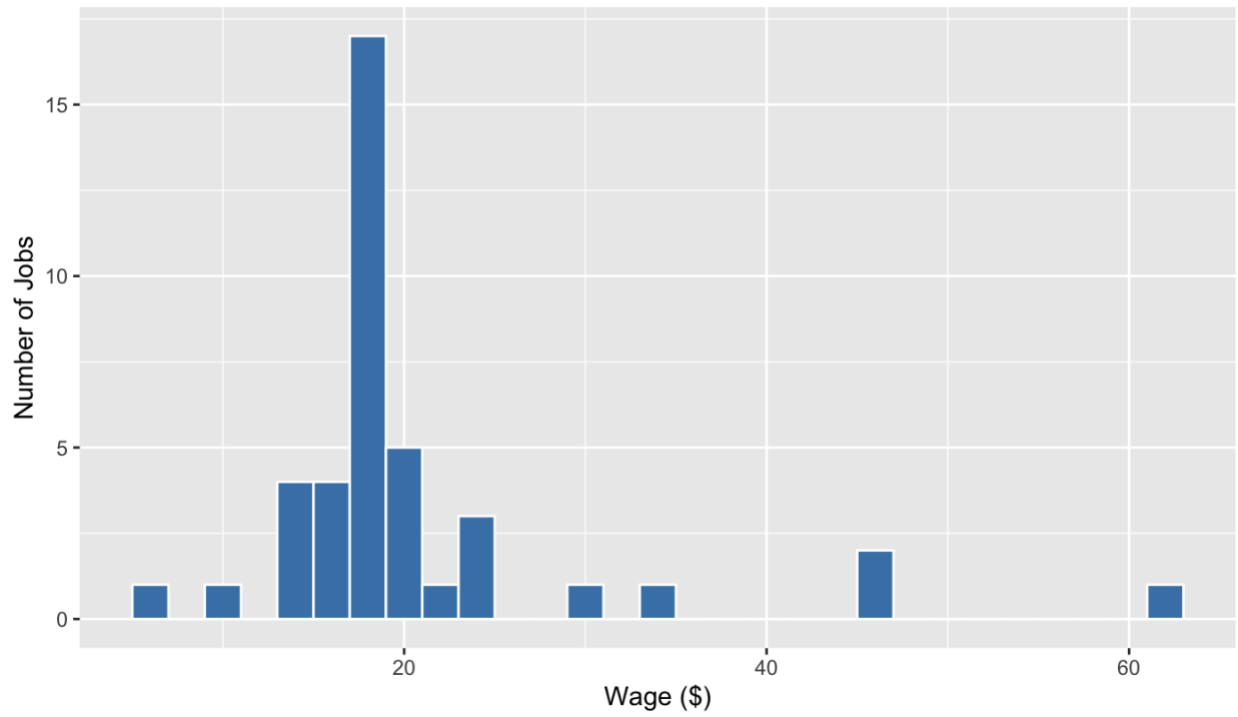
Program career occupation title vs same title on O*Net Online

Career Occupation Title	O*Net Online Occupation Name
Teacher Support	Teacher Assistants, Special Education
Probation Officer	Probation Officers and Correctional Treatment Specialists
Licensed Practical Nurse	Licensed Practical and Licensed Vocational Nurses
IT Specialist	Computer User Support Specialists
Human Resources Manager	Human Resources Managers
Entrepreneur	Chief Executives
Emergency Medical Technician	Emergency Medical Technicians
Customer Service Representative	Customer Service Representatives
Chef	Chefs and Head Cooks
CDL Bus Driver	Bus Drivers, Transit and Intercity
Administrative Support	Office and Administrative Support Workers,

	All Other
Mail Carrier	Postal Service Mail Carriers

Career Goal Categories and Keyword Patterns	
Career Category	Pattern Keywords
Social Work	social case human services custodial community outreach
Dental Hygienist	dental hygienist pharmacy technician
Nursing	nurs surgical phlebot sonograph respiratory radiology midwife therapist mental health scientist labor/delivery physician veterinarian
Healthcare Support	assistant aide medical coder billing veterinary health information management
Education	teacher educat paraprofessional school
Childcare	child cda
Entrepreneurship	entrepreneur owner salon cater business entrepeneur enterpreneur
Financial Services	finance fundraising account insurance payroll financial advisor financial hr professional
Peer Support	peer mental health
Maintenance	plumb maintenance custodial technician environmental housekeep dishwasher maintenace worker waste management
Legal	attorney paralegal judge
Real Estate	real estate realtor leasing architectural
IT Professionals	it cyber technician microsoft google comptia cisco
Customer Service	call center customer service dispatcher receptionist tsa
Law Enforcement	mpd police law enforce marshal security officer crime sergeant superintendent
Security Guards	security guard armed security unarmed security protective officer
Administrative	admin clerical office human resource
Chef/ Food Related	chef cook server restaurant food baker
Program Management	program specialist program manager director contractor
Drivers	cdl driver bus operator truck motor vehicle operator
Cosmetology	cosmetolog esthetician barber massage groomer
Retail Management	manager director supervisor sales associate management
Federal/ Govt Work	usps federal fema gov
Hospitality	hotel hospitality funeral
Special Fields	engineer firefighter vendor machine operator
Undecided	undecided ^na\$ aspiring instructor

Distribution of Hourly Wages



Distribution of Desired Career Interests

